

Internship Project Report On

A Project based On GATE PASS MANAGEMENT SYSTEM

A report submitted as a part of the industrial Orientation Training in IT & ERP Department.

Visakhapatnam Steel Plant.

By

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RASTRIYA ISPAT NIGAM LIMITED (RINL)

Visakhapatnam Steel Plant

(Duration: 19th May 2025 to 14th June 2025)

CERTIFICATE

This is to certify that the allowing Students of **MAHARAJA VIJAYARAM GAJAPATHI RAJ COLLEGE OF ENGINEERING** are engaged in the project work sided A Project Band Gate pass management system from 19th May 2025 to 14th June 2025.

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In partial Fulfilment of the degree **BACHELORS OF TECHNOLOGY** in **COMPUTER SCIENCE ENGINEERING & INFORMATION TECHONOLOGY** stream in **MAHARAJA VIJAYARAM GAJAPATHI RAJ COLLEGE OF ENGINEERING** is a record of Bonafide work carried out by an under the guidance and supervision during the period from 19th may 2025 to 14th June 2025

Date:

Place: Visakhapatnam

V. HEMA SUNDAR RAO

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ACKNOWLEDGEMENT

First of all, I thank God in words for his grace, who gave us the opportunity and strength to carry out this work. The success and outcome of this project required a lot of guidance and assistance from many people and we are incredibly privileged to have got this all along with the completion of our project.

I thank to having this opportunity to thank Mr. V.H. SUNDHARA RAO, IT and ERP Department. Visakhapatnam Steel Plant for guiding us with his immense knowledge and helping us to complete this project successfully.

I wish to express our sincere thanks to IT and ERP Dept. employees of Visakhapatnam Steel Plant for their valuable guidance in completing this project in time Successfully.

DECLARATION

I hereby that this project work entitled A Project Based on **Gate pass management system** is original and has not been submitted to any university or college before fulfilment of the requirements of any course of study or to the award of any degree. The opinion is given and the conclusions arrived at are of my own. The views expressed in the report do not represent the views of organization.

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ABSTRACT

The Gate Pass Management System is a digital solution designed to streamline and automate the process of issuing, tracking, and managing gate passes within an organization, such as educational institutions, corporate offices, or industrial facilities. Traditionally, gate pass management involves manual entries and physical paperwork, which can lead to inefficiencies, data loss, and security vulnerabilities. This system aims to replace the manual approach with a secure, centralized, and user-friendly web or mobile-based application.

The system facilitates the generation of gate passes for various purposes—student or employee movement, visitor entry, or goods dispatch—ensuring all necessary approvals are obtained electronically. Key features include user authentication, role-based access control, real-time status tracking, automated notifications to stakeholders, and a searchable record of past passes. By implementing this system, organizations can enhance security, reduce paperwork, improve accountability, and maintain accurate records with minimal human intervention.

This project is developed using modern technologies such as [specify tech stack if required: e.g., HTML, CSS, JavaScript, PHP, MySQL], ensuring scalability, reliability, and ease of integration with existing infrastructure.

INTRODUCTION TO VSP

A BRIEF OVERVIEW:

Visakhapatnam Steel Plant (VSP) is the integrated steel plant Rashtriya Ispat Nigam Limited in Visakhapatnam which is founded in 1971, VSP strikes every one with a tremendous sense of awe, wonder and amazement as it presents a wide array of excellence in all its facets in scenic beauty, in management and above all in product quality. On the coast of Bay of Bengal and by the side of scenic Gangavaram beach, have risen tall and huge structures of technological architecture, the Visakhapatnam Steel Plant. But the vistas of excellence do not rest with the inherent beauty of location over sophistication of technology- they march ahead parading one aspect after another.

The decision of the Government of India to set up an integrated steel plant at Visakhapatnam was announced by then Prime Minister Smt. Indira Gandhi in Parliament on 17th January 1971. VSP is the first coastal based integrated steel plant of India is located, 16km west of city of destiny, Visakhapatnam, bestowed with modern technologies, VSP has an installed capacity of 3 million tonnes per annum of liquid steel and 2.656 million tonnes of saleable steel. The saleable steel here is in the form of Wire rod coils, Structural, Special Steel, Rebars, Forged

Rounds and etc. At VSP there emphasis on total automation, seamless integration and efficient up gradation, which result in wide range of long and structural products to meet stringent demands of discerning customers with India & abroad. VSP product meet exalting international Quality Standards such as JIS, DIN, BIS, BS etc. Rashtriya Ispat Nigam Limited, the corporate entity of Visakhapatnam Steel Plant is a Navaratna Public Sector Enterprise under the Ministry of Steel. Visakhapatnam Steel Plant is also fondly called as Vizag Steel. It is the first shore based Integrated Steel Plant in the country and is known for it's quality products delighting the customers. It is market leader in long products and it caters to the needs of diverse industrial sectors. It is the first steel plant to be certified ISO9001:2008 (presently 2015), ISO 14001:2004(presently 2015), OHSAS 18001:2007 and ISO/IEC 27001:2013 Standards. It is also the first PSE to be certified ISO 50001:2011- Energy Management Systems and has acquired CMMI Level 3 certification for S/W development. On 20th Jan 71 the then Prime minister of India laid the foundation stone of this plant. A new company Rashtriya Ispat Nigam Limited (RINL) was formed on 18th Feb 1982. From 1987 to 1992 various units of VSP were commissioned. VSP's Product Mix comprises Wire Rods, Bars, Angles, Channels/Beams, Rounds and Billets. The Plant also produces Pig Iron, Granulated Slagand Coal Chemicals. The rolled products find extensive usage in the Construction, Infrastructure, Railways, Power, Defence. Transport and Ship Building sectors. Coir's and Rods are used mainly for

reinforced concrete work for housing, construction of dams, buildings factories, manufacture of agricultural implements, fabrication of light engineering components. The Wire rods are used in Wire Drawing industry for electrodes transmission lines etc. The structural find application in engineering, house building, agricultural implements machinery, transmission towers, etc. VSP is having the state of the art technology.

The steel plant has many technological features, which are unique amongst the steel plants in the country. The company is a pioneer in introducing many new technologies in the country. The production of TMT rebars by temporal process is a shining example in this respect.

VSP has become the first integrated steel plant in the country to be certified to all the three international standards for quality (ISO-9001), for Environment Management (ISO) -14001) & for Occupational Health & Safety (OHSAS 18001). The certificate covers quality systems of all operational, maintenance, service units besides purchase systems, Training and Marketing functions spreading over 4 Regional Marketing offices, 20 branch offices & 22 stock yards located all over the country. VSP by successfully installing & operating efficiently Rs. 460 process worth of pollution and Environment Control Equipment and converting the barren landscape more than 3 million plants has made the steel plant, steel Township and VSP exports Quality Pig Iron & Steel products Sri Lanka, Myanmar, Nepal, Middle East, USA & South East Asia (Pig Iron). RINL-VSP was awarded "Star

Trading House" status during 1997-2000. Having established a fairly dependable export market, VSP plans to make a continuous presence in the export market.

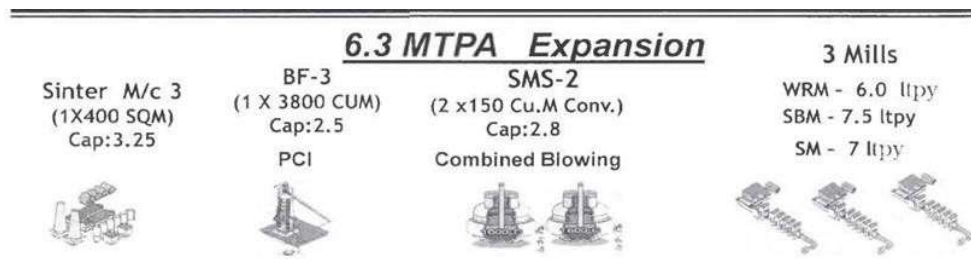
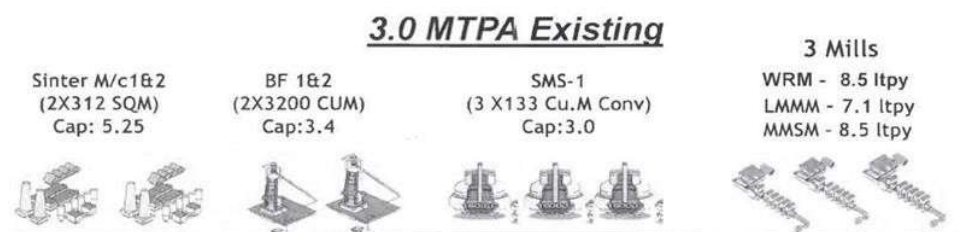
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Different sections at the RINL VSP:

- Coke oven and coal chemicals plant
- Sinter plant
- Blast Furnace
- Steel Melt Shop
- Continuous casting machine
- Light and medium machine mills
- Calcining and refractive materials plant
- Rolling mills

- Thermal power plant Chemical power plant
- Yard management

The expansion of production facilities from 3MTPA stage to 7.3 MTPA stage has been shown below:



SINTER PLANT

Sinter plants agglomerate iron ore fines (dust) with other fine materials at high temperature, to create a product that can be used in a blast furnace. The final product, a sinter, is a small, irregular nodule of iron mixed with small amounts of other minerals. The process, called sintering, causes the constituent materials to fuse to make a single porous mass with little change in the chemical properties of the ingredients. The purpose of sinter are to be used converting iron into steel.

Sinter plants, in combination with blast furnaces, are also used in non-ferrous smelting. About 70% of the world's primary lead production is still produced using the sinter plant–blast furnace combination,[1] and this combination was formerly often used in copper smelting (at the Electrolytic Refining and Smelting smelter in Wollongong, New South Wales,

INFORMATION TECHNOLOGY & ERP DEPARTMENT

VSP is the first Steel Plant in India where integrated Automation and computerization is envisaged right from the beginning. Almost all the production and business processes are controlled by computer in one form or the other. These computers and the related automation systems are meant to aid all levels of personnel involved in operation and maintenance of the plant. They provide valuable information for the top management to take both long and short-term decisions. IT applications at RINL have been developed and implemented keeping the overall organizational business objectives in view. IT infrastructure has been upgraded recently with IBM-RS 6000 servers and ORACLE-91 as the database server and data communication with backbone Fiber- optic network. In the area of marketing VPN based wide area networks has been implemented to provide vital information to regional marketing offices as well as customers. **VSP** has a sizable amount of computerization built into its design in line with the latest trends in technology. The main objective of computerization in **VSP** is to assist the operators in controlling repetitive jobs, to optimize utilization of scarce resources by real time process control and assist in operation and production management. The computerization and automation in **VSP** is envisaged in three levels for effective integration between planning and processes. The highest level assists management decision-

making by collection, analysis and providing the production and management information on a real time basis and the lowest level assists the actual process operators by reducing their tedium in the form of automation. Thus the structure is designed for assisting all levels of operation and management of the plant.

BUSINESS APPLICATION SYSTEMS:

This is for top level of the hierarchy and is consisting of a Central Computer. All the large online business applications covering different business functions of the plant are implemented in this computer. This generates all the information/reports for the management control. It also stores and provides long-range plant operating, maintenance, commercial and finance data. Based on the long-range information this computer plans the production scheduling for the shops. It generates data regarding the products to be produced, maintenance plans and dispatch plans. These production and maintenance plans are transmitted to the concerned process control computers/personnel. Subsequently the actual plant operational, maintenance data are collected. Some of the important business applications running in VSP are listed below:

1. Payroll System
2. Costing System

3. Financial Accounting System
4. Marketing System
5. Attendance Recording System
6. Library Information System
7. Materials Management System
8. Maintenance Management System
9. Production Planning and Control System
10. Wagon Information Tracking System
11. On-line TC printing System
12. Human Resource Information System
13. Raw Material Management System
14. Coke Ovens Heating and Regulating

System PROCESS CONTROL SYSTEMS :

The main jobs of the process control computers are memorizing production schedules given by zonal computers, monitoring the shop production and the material flow in the shops and storing and execution of process control parameters and algorithms to control the process. They also control the lower level automation systems. The most important aspect of these computers is the man machine dialogue envisages. Such

computer systems are envisaged in the Blast Furnaces, Steel Melt Shop, Continuous Casting Machines & Rolling Mills. Apart from these computers, there is a 'Supervisory Control And Data Acquisition (SCADA)' computer to monitor and control Gas, Power, Water and their distribution in the plant.

FIELD LEVEL AUTOMATION SYSTEMS:

This level consists of local automation systems. These systems do logic, sequence, loop control and indication of process parameters to operation personnel in the plant for taking up timely corrective actions to achieve the quality production. This level consists of Programmable Logic Control (PLC) Systems, analog (split architecture type) and microprocessor-based instrumentation systems. The above concept of hierarchical and integrated computer systems in line with the latest technology is introduced in VSP for the first time in India. This system is contributing a lot in achieving the quality and productivity of steel with an integrated control of production and processes. Thus, VSP is a trendsetter in computerization in basic industries in India with latest systems and opening new vistas in computerization in our country.

INTRANET WEB SITE :

An intranet web site is also maintained by VSP IT Department and many facilities are hosted in this site for the convenience of the different sections of the plant and for the benefit of the employees. The content in this site is

updated regularly. There are many important features/modules hosted in this site.

NETWORK

VSP is having a Gigabit Ethernet Data Communication Network, working on Optical Fiber Cable(OFC) network. This supports data transfer at the rate of 1000 Mbps on all the main links. The campus network is divided into eleven zones. These zones are inter-linked to the central core switch using Single mode type optical fibers. The Central core switch is located at Central Computer Centre. The 11 Zonal switches are located at Coke Ovens, Sinter Plant, SMS, Central stores, BC Gate, LMMM, MMSM, Central computer centre, Admin building, Township and General Hospital. Further, a total of 200 access switches are connected to the network through the 11 zonal switches, using Multimode/ single mode types fiber optical cable,. The access switches provide connectivity to approximately 3000 nodes throughout the plant premises, Administration building area and township. ERP It stands for Enterprise Resource Planning. ERP refers to a suite of software that organizations use to manage day-to-day business activities, such as accounting, procurement, project management, risk management, compliance, and supply chain operations. In Visakhapatnam Steel Plant ERP is being used extensively and it is covering many applications and most of the departments.

INTRODUCTION TO THE PROJECT

In many organizations, controlling and monitoring the movement of individuals and materials in and out of the premises is crucial for ensuring security and operational efficiency. The traditional method of managing gate passes—often reliant on paper-based records—is time-consuming, prone to human error, and lacks real-time tracking and accountability. To overcome these limitations, a **Gate Pass Management System** is proposed.

This system is a digital platform that automates the gate pass process for students, staff, visitors, and goods. It allows users to request gate passes online, and enables authorized personnel to approve, reject, or track those requests efficiently. The system maintains a secure database of all gate pass records, which can be easily searched and audited. It also enhances security by ensuring that no unauthorized entries or exits take place without approval and logging.

We have created a website for automatically storing the log sheet values of the Yard Management Dispatch Process System in the database using-

- 1) Frontend (HTML, CSS, JavaScript)
- 2) Backend (Node.js, Express.js, MYSQL)

HTML-

The Hypertext Markup Language, or HTML is the Standard Markup Language for documents designed to be displaying in a web browser. It can be assisted by technologies such as Cascading Style Sheets (CSS) and scripting languages as JavaScript. Web Browsers receive HTML documents from a webserver or from local storage and render the documents into multimedia webpages. HTML describes the structure of a webpage semantically and originally includes cues for appearance of document.

CSS-

Cascading Style Sheets (CSS) is a style sheet language used for describing the presentation of a document written in a markup language such as HTML. CSS is a corner stone technology of the World Wide Web, alongside HTML and JavaScript. CSS is designed to enable the separation of presentation and content, including layout, colors and fonts. This separation can improve content accessibility; provide more flexibility and control in the specification of presentation characteristics; enable multiple webpages to share formatting by specifying the relevant CSS in a separate .css file, which reduces complexity and repetition in the structural content; and enable the .css file to be cached to improve the page load speed between the pages that share the file and its formatting.

JAVASCRIPT-

JavaScript often abbreviated JS, is a programming language that is one of the core technologies of the World Wide Web, alongside HTML and CSS. Over 97% of websites use JavaScript on the client side for web page behavior, often

incorporating third-party libraries. Major web browsers have a dedicated JavaScript engine to execute the code on user devices.

NODE JS-

Node.js is a powerful, open-source JavaScript runtime built on Chrome's V8 engine. It allows developers to run JavaScript code outside of a browser—typically on servers—making it ideal for building fast, scalable, and real-time applications like web APIs, chat apps, and streaming services. Node.js uses an event-driven, non-blocking I/O model, which helps handle multiple connections efficiently.

EXPRESS JS-

Express.js is a lightweight and fast web framework built on top of Node.js. It helps developers create web applications and APIs more easily by handling tasks like routing (directing URLs to the correct code), managing requests and responses, and supporting middleware (code that runs before a request reaches the final handler). Instead of writing complex server code from scratch, Express simplifies it with easy-to-use functions and a clean structure.

It's commonly used for building RESTful APIs and full-stack web apps, and works well with frontend frameworks like React, Angular, or plain HTML/CSS.

MYSQL-

MySQL is currently the most popular database management system software used for managing the relational database. It is open-source database software, which is supported by Oracle Company. It is fast, scalable, and easy to

use database management system in comparison with Microsoft SQL Server and Oracle Database. It is commonly used in conjunction with PHP scripts for creating powerful and dynamic server- side or web-based enterprise applications.

It is developed, marketed, and supported by MySQL AB, a Swedish company, and written in C programming language and C++ programming language. The official pronunciation of MySQL is not the My Sequel; it is My Ess Que Ell. However, you can pronounce it in your way. Many small and big companies use MySQL. MySQL supports many Operating Systems like Windows, Linux, MacOS, etc. with C, C++, and Java languages.

MySQL is a Relational Database Management System (RDBMS) software that provides many things, which are as follows:

- It allows us to implement database operations on tables, rows, columns, and indexes.
- It defines the database relationship in the form of tables (collection of rows and columns), also known as relations.
- It provides the Referential Integrity between rows or columns of various tables.
- It allows us to updates the table indexes automatically.
- It uses many SQL queries and combines useful information from multiple tables for the end-users

DESIGN OF PROJECT AND DATABASE SCHEMA

The below is the project overview and the design which was made before writing and developing the code for the website.

Project Modules Overview

This portal is divided into five core modules:

- **User Authentication Module**
- **Worker Management Module**
- **Attendance Tracking Module**
- **Attendance Reports and Analytics**
- **Session Tracking Module (Proposed)**

UI/UX Design

UI/UX Design refers to two distinct but deeply intertwined disciplines that focus on creating meaningful and user-friendly digital products (like websites, mobile apps, software) and sometimes even physical products or services. They both aim to enhance the user's satisfaction and experience.

breakdown of each:

UX Design (User Experience Design)

Definition:

User Experience (UX) design is the process of enhancing user satisfaction by improving the usability, accessibility, and pleasure provided in the interaction between a user and a product or service. It's about understanding what users need, what their goals are, and designing a product that effectively helps them achieve those goals while making the experience as intuitive and enjoyable as possible.

Description:

Think of UX design as the **architectural blueprint** and the **invisible backbone** of a product. A UX designer is concerned with the *entire journey* a user takes when interacting with something. This includes:

- **Research:** Deeply understanding the target users through interviews, surveys, and analyzing their behaviors, needs, and pain points. They ask questions like: Who are the users? What are their motivations? What problems are they trying to solve?
- **Information Architecture (IA):** Organizing and structuring content in a logical and easy-to-find way. This is like creating the sitemap for a website or the navigation flow for an app, ensuring users can find what they're looking for intuitively.

- **Interaction Design (IxD):** Defining how users will interact with the product – what happens when they click a button, swipe, or enter text. It's about designing the "flow" and responsiveness of the system.
- **Wireframing and Prototyping:** Creating low-fidelity (basic outlines) and high-fidelity (more detailed, interactive models) representations of the product to test flows and gather early feedback before visual design is applied.
- **Usability Testing:** Observing real users interacting with prototypes or live products to identify areas of confusion, frustration, or inefficiency, and then iterating on the design based on that feedback.
- **Accessibility:** Ensuring the product can be used by people with diverse abilities and disabilities, adhering to guidelines that make it inclusive for everyone.

In essence, UX design is about the "why," "what," and "how" of product use. It focuses on solving user problems and ensuring the product is useful, usable, desirable, and valuable.

UI Design (User Interface Design)

Definition:

User Interface (UI) design is the process of designing the visual and interactive elements of a digital product's interface. It focuses on the "look and feel" – how the product is presented to the user, including the screens, buttons, icons,

typography, colors, and other visual components that users directly interact with.

Description:

If UX is the blueprint, then UI is the **interior design** and **decoration** of the product. A UI designer takes the structural and functional framework laid out by the UX designer and brings it to life visually. Their responsibilities include:

- **Visual Design:** Choosing and applying color palettes, typography, imagery, icons, and graphical elements to create a consistent and aesthetically pleasing visual identity for the product.
- **Layout and Grids:** Arranging all the visual elements on the screen in an organized and harmonious way, considering spacing, alignment, and visual hierarchy to guide the user's eye.
- **Interactive Elements:** Designing specific UI components like buttons, sliders, input fields, toggles, menus, and notifications, ensuring they are not only visually appealing but also clear, intuitive, and consistent in their behavior.
- **Branding and Consistency:** Ensuring that the visual design reflects the brand's identity and that design elements are consistent across all screens and platforms for a unified user experience.
- **Responsive Design:** Designing interfaces that adapt and look good on various screen sizes and devices (desktops, tablets, mobile phones).
- **Micro-interactions and Animations:** Adding subtle animations,

transitions, and feedback elements that make the interface feel alive, responsive, and delightful to use.

In essence, UI design is about the "aesthetics" and "interactivity" of the product. It focuses on making the product visually appealing, easy to navigate, and pleasurable to interact with.

The Relationship Between UI and UX:

While distinct, UI and UX are interdependent. A fantastic user experience can be undermined by a poor user interface, and a beautiful interface won't save a product that's difficult to use or doesn't meet user needs.

- UX comes first: Typically, UX research and strategy inform UI design. You figure out *what* needs to be built and *how* it should function (UX) before designing *how it looks* (UI).
- UI brings UX to life: The UI is the tangible manifestation of the UX. A well-designed UI makes the well-thought-out UX accessible and enjoyable.
- They collaborate: UI and UX designers often work closely together in a team, with the UX designer providing the foundational research and wireframes, and the UI designer crafting the visual and interactive details based on that foundation.

Together, UI/UX Design creates products that are not only functional and useful but also delightful and easy to use, leading to increased user satisfaction

Consistent Layout: Sidebar navigation, top bar with branding

Responsive Design: Mobile-friendly UI, table scrolling support

Accessible Colors: High contrast for readability

Loading Indicators: For data fetch operations

User Authentication Module

- A User Authentication Module is a critical component in almost any software application or system that needs to restrict access to certain information or functionalities. It's essentially the gatekeeper that verifies who you are before letting you in.
- A User Authentication Module (or simply "Authentication Module") is a **software component or system designed to verify the identity of a user attempting to access a digital system, application, or resource.** Its primary purpose is to ensure that only legitimate and authorized individuals can gain entry.

Purpose:

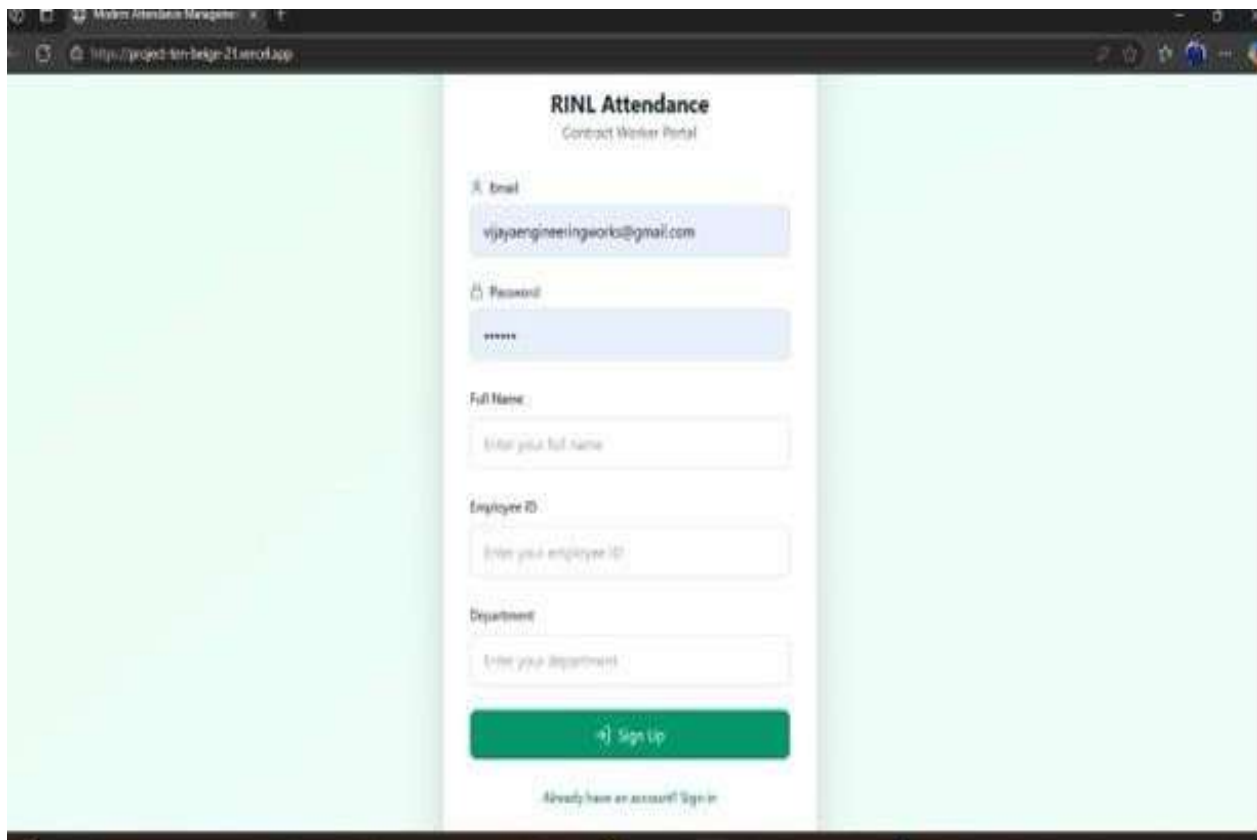
Ensures only authorized personnel (HR staff or department users) can access and manage the portal.

Features:

Sign Up: fig:1

Required fields:

- Full Name
- Department Name
- Email Address
- Password (with validations)
- Employee ID
- Verifies uniqueness of email or ID before account creation.



The screenshot shows a web browser window with the URL <http://project-tin-hai-21.herokuapp.com>. The page title is "RINL Attendance" and the subtitle is "Contract Worker Portal". The form contains the following fields and elements:

- Email:** A text input field containing the email address "vijayaengineeringworks@gmail.com".
- Password:** A password input field with masked characters "*****".
- Full Name:** A text input field with the placeholder text "Enter your full name".
- Employee ID:** A text input field with the placeholder text "Enter your employee ID".
- Department:** A text input field with the placeholder text "Enter your department".
- Sign Up:** A green button with a right-pointing arrow icon and the text "Sign Up".
- Link:** A link at the bottom that says "Already have an account? Sign in".

Fig:1

Login: fig:2

Accepts registered email & password.

Redirects to the dashboard upon success.

Invalid credentials display relevant error messages.

Security:

Frontend validations for input sanitization.

Backend token/session-based authentication (JWT or Firebase Auth assumed).

Passwords securely hashed (if backend uses own auth system).

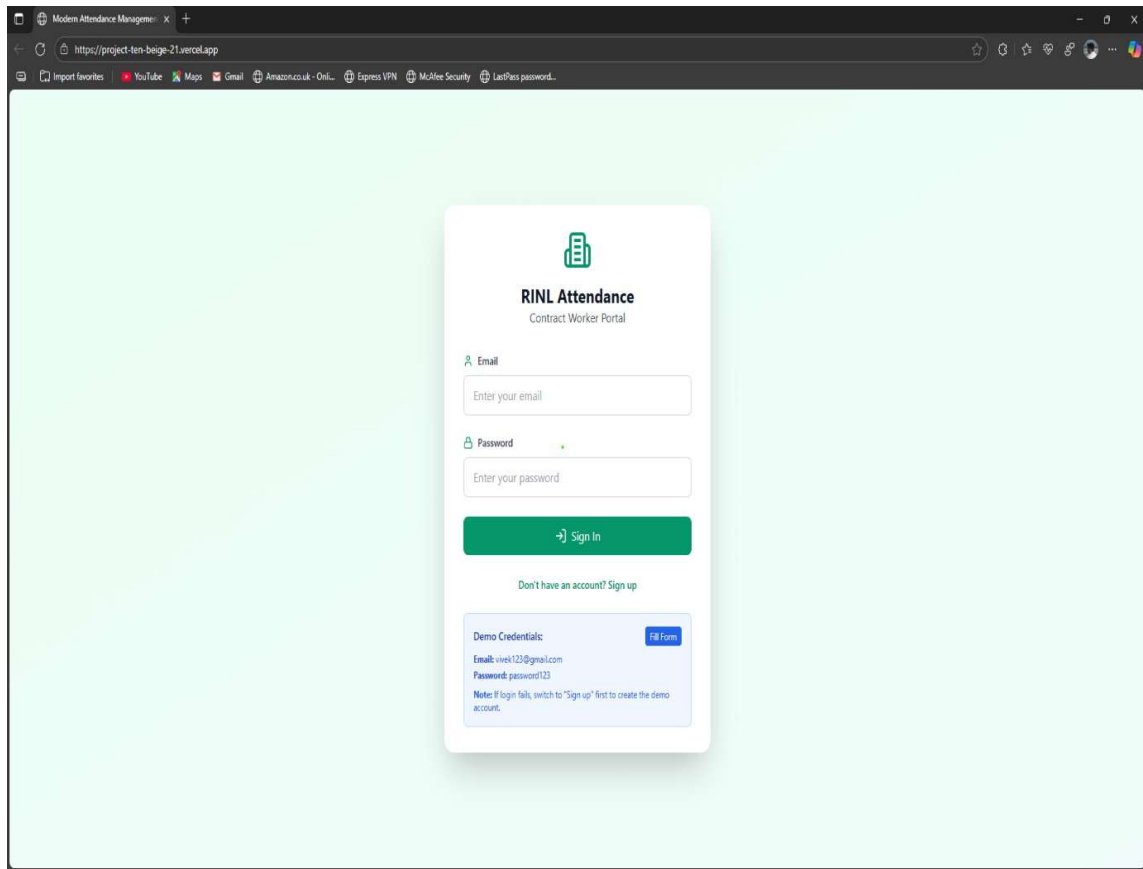


Fig:2

Worker Management Module

Purpose:

Handles onboarding and managing records of contract workers employed under various RINL departments.

Interface Overview:

List View of Workers:

Columns include:

- Name
- Aadhaar Number
- Skill Level (SK / SSK / USK)
- Gate Pass Number
- Pass Expiry Date

Actions:

-  Edit
-  Delete

Add Worker Form: fig:3

Inputs:

- Name
- Aadhaar Number (12-digit validation)
- Skill Category
- Gate Pass Number
- Gate Pass Validity

The screenshot displays the 'RINI Contract Worker Portal' interface. The header includes the portal name and a 'Sign Out' button. Below the header, there are navigation links for 'Worker Management', 'Mark Attendance', 'Reports', and 'Session Tracking'. The main content area is titled 'Worker Management' and features a search bar and an 'Add Worker' button. A table lists several workers with their details and actions.

NAME	AADHAAR ID	SEX	DATE OF BIRTH	ACTIONS
THEMBURU GIRIDASI	9.695196+11	SE	-	
PASUPUREDDY MURALI	6.888623+11	SM	-	
YENNA KANAKA RAJU	3.888340+11	SM	-	
RAMAKRISHNA CHIDAMARU	4.530233+11	SM	-	
PUDI APRNAD	5.554736+11	SM	-	
VANGAMATI NAGESWARARAO	4.179333+11	SM	-	
...	...	...	...	...

Total Workers: 10 | Showing: 22

Fig:3

Attendance Tracking Module

Purpose:

Enables departments to record daily attendance of contract workers efficiently.

Features:

- Displays list of registered workers.
- Checkboxes to mark presence per worker for selected date.
- Automatic timestamping of attendance data.
- Restricts duplicate entries for the same date.

User Experience:

- Minimalist and responsive UI.
- Quick batch attendance marking.
- Visual confirmation after submission

Modern Attendance Management

https://project-kms-beta-21.mvclapp

RINL Contract Worker Portal

Welding, Rigging Engineering works (P. & Up Department)

[+ Sign Out]

Worker Management | **Mark Attendance** | Reports | Session Tracking

Attendance Date: 11-06-2025

Shift: Shift A

[Save Attendance]

Select Workers for Attendance

[Select All] [Unselect All]

☒ ATTILU HOOKARAJU SR 637134-11	☒ BILLINGI PRAKASARAO SR 553455-11	☐ DUVVU JAGGARAO SR 634466-11
☐ GALLA LOVARAJU SR 120222-11	☐ GALLA RAMESH SR 620008-11	☐ GODDI GOVINDU RAJ SR 632203-11
☐ innocent one SR 67000075407	☐ KARGALA HARI BABU SR 640075-11	☐ LAKKARAJU RAVI KUMAR SR 640754-11

Fig:4

Attendance Reports & Analytics Module

Purpose:

To view, analyse, and export attendance trends over customizable time ranges.

Features:

- Date Range Selection:
Start Date & End Date (via calendar picker)
- Worker Filter:
All workers or specific worker
- Stats Display:
Total Workers

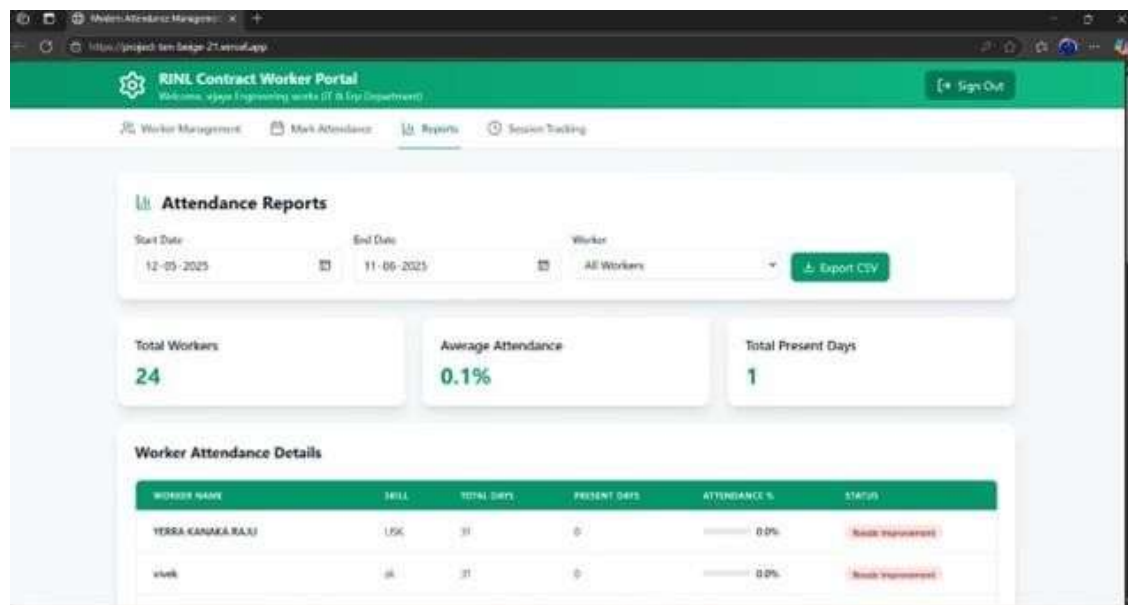


Fig:5

Average Attendance (%)

- Total Days Tracked
- Total Present Days

Attendance Table:

Worker Name	Skill	Total Days	Present Days	Attendance %	Status
Rajesh K	SK	25	21	84%	Satisfactory
Ali B	USK	25	15	60%	Need Attention

Export CSV:

- One-click CSV download of report data
- Useful for audits and offline reporting

Session Tracking Module (Planned / Placeholder)**Purpose:**

To monitor work hours or session durations for each worker per day.

Potential Use-Cases:

- Clock-in and Clock-out logs
- Total working hours calculation
- Overtime tracking

Status:

Present as a navigation item in UI

Functionality not fully implemented as of now

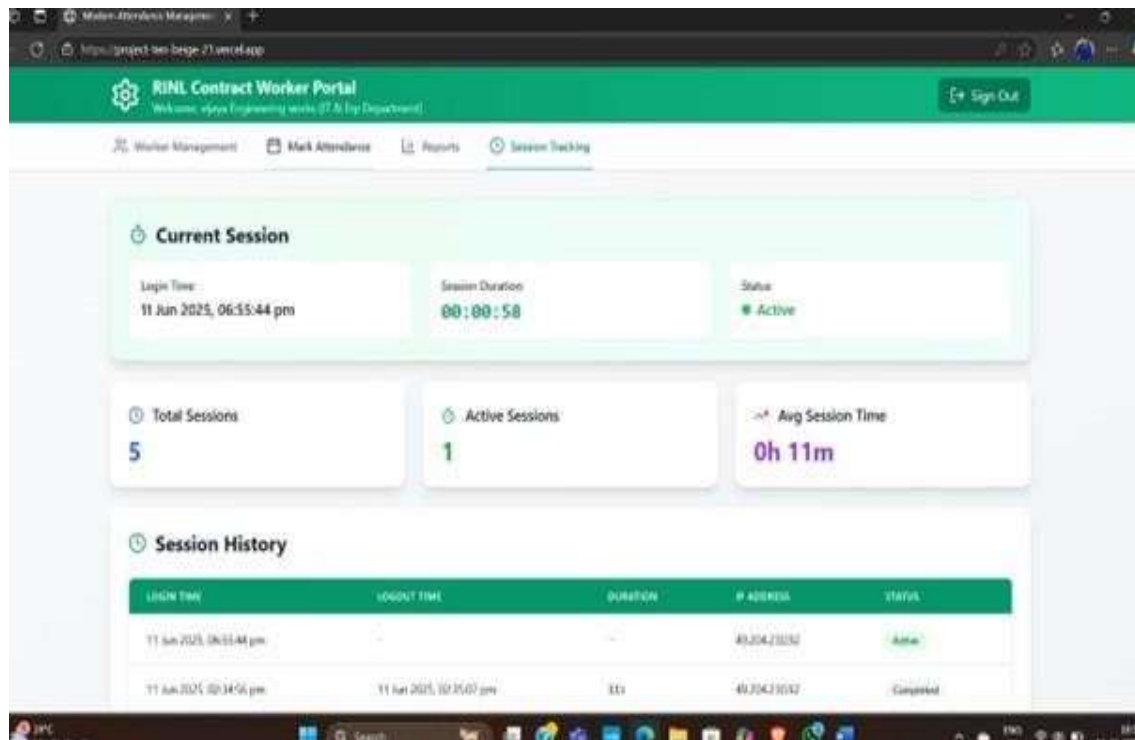
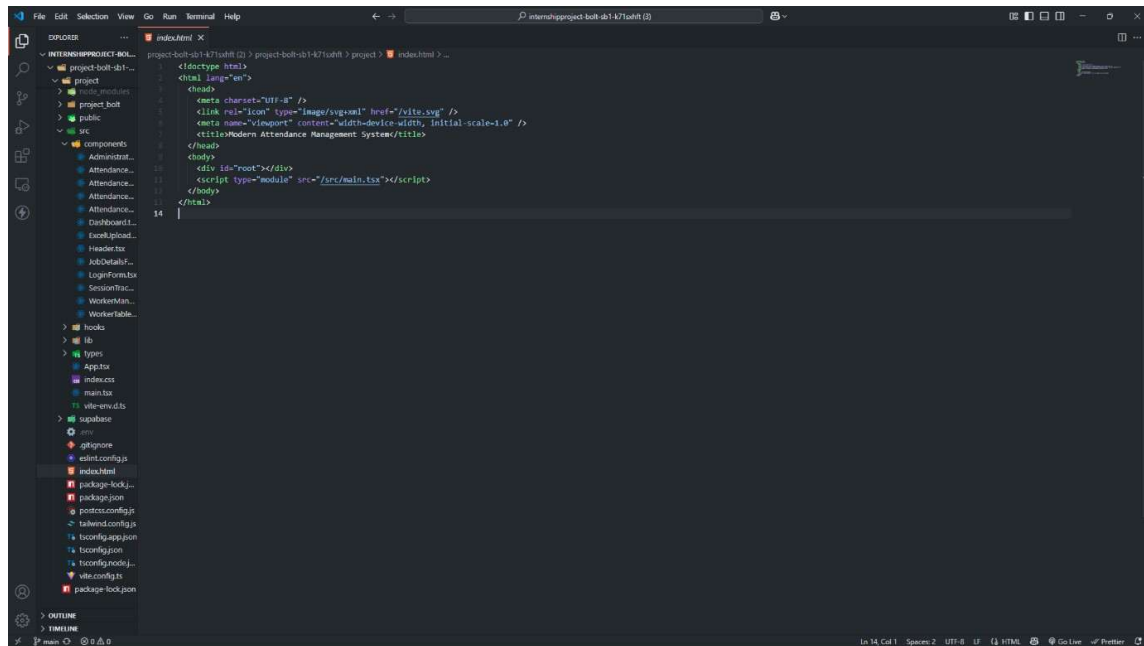


Fig:6

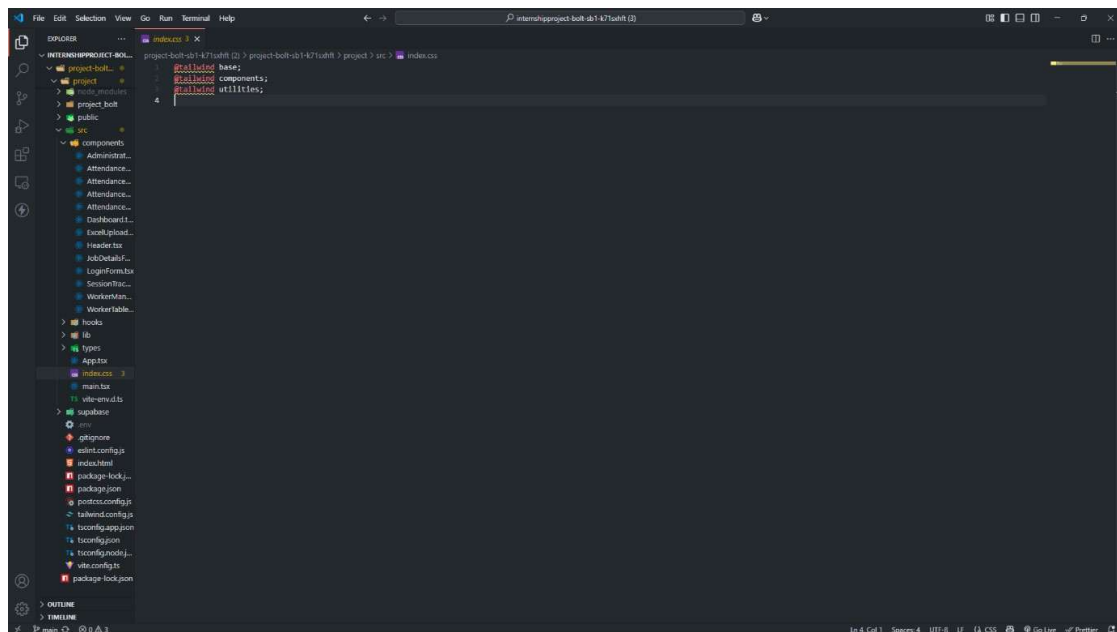
Sample code

HTML:



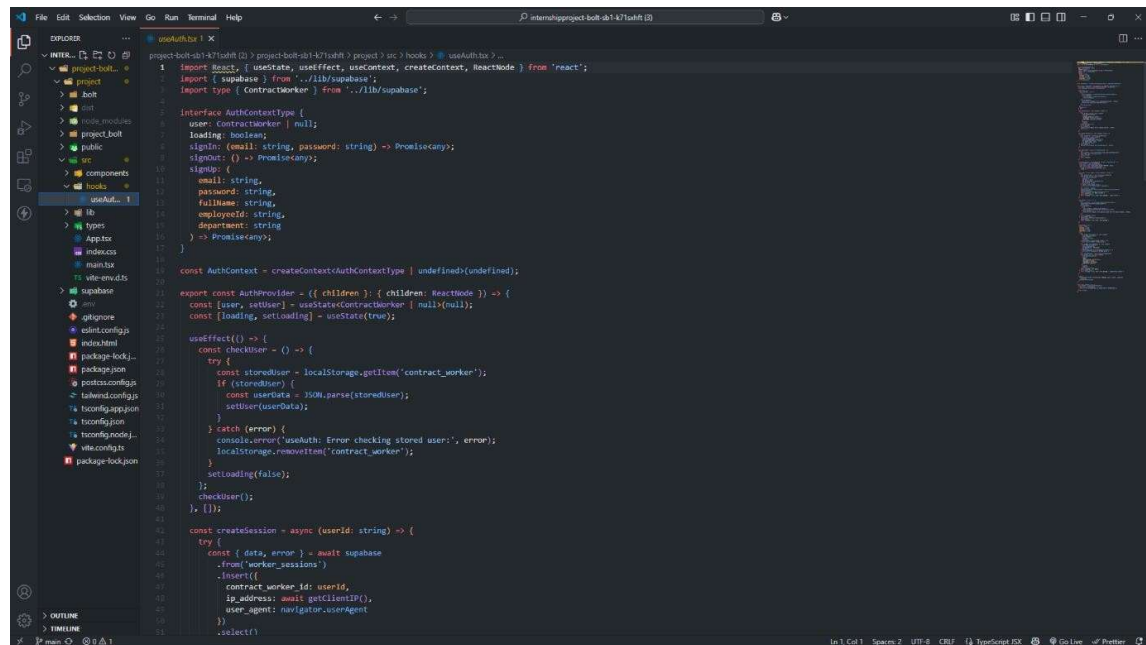
```
index.html X
project-bolt-d1-k7tsuht (2) > project-bolt-d1-k7tsuht > project > index.html > ...
1 <!doctype html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <link rel="icon" type="image/svg+xml" href="/vite.svg" />
6   <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7   <title>Modern Attendance Management System</title>
8 </head>
9 <body>
10   <div id="root"></div>
11   <script type="module" src="/src/main.tsx"></script>
12 </body>
13 </html>
14
```

CSS:



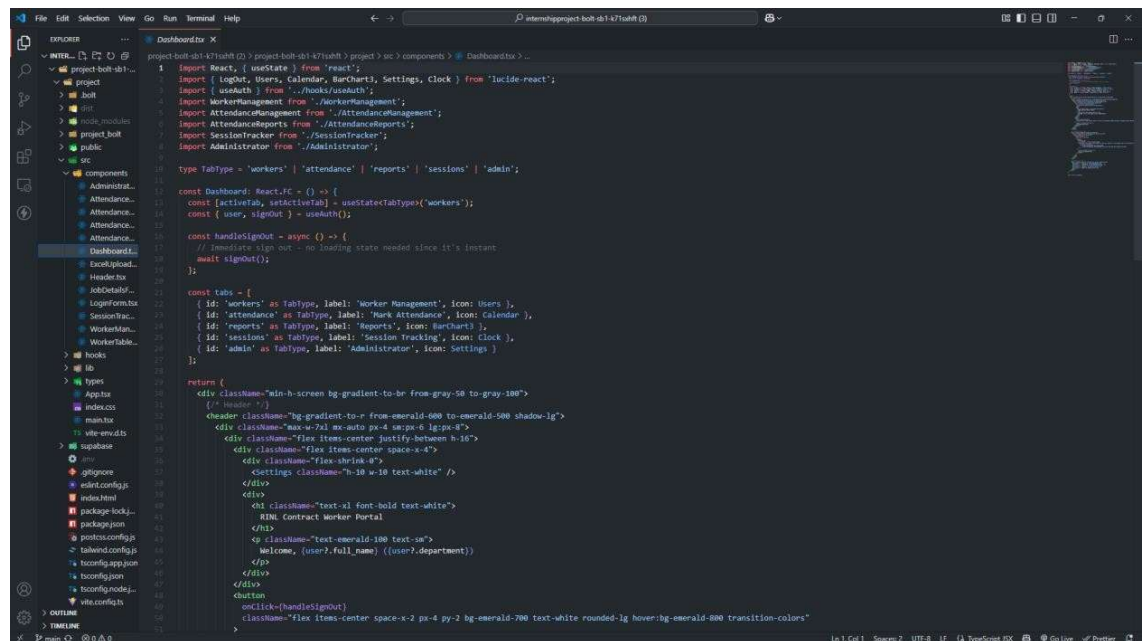
```
index.css X
project-bolt-d1-k7tsuht (2) > project-bolt-d1-k7tsuht > project > src > index.css
1 @tailwind base;
2 @tailwind components;
3 @tailwind utilities;
4
```

User Authentication:



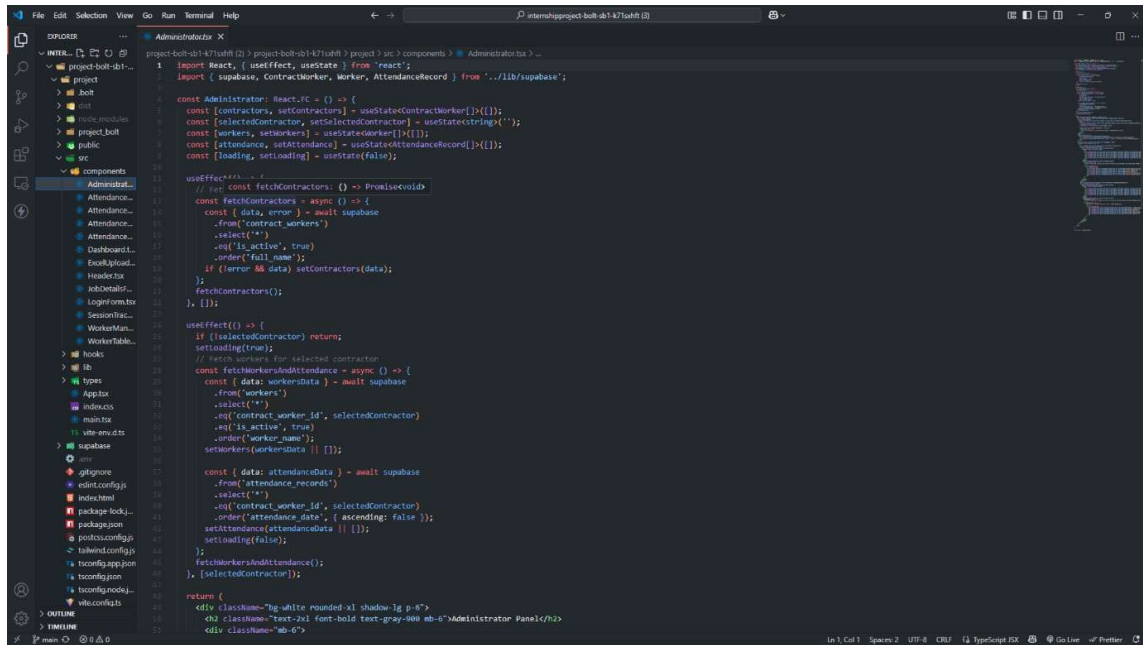
```
1 import React, { useState, useEffect, useContext, createContext, ReactNode } from 'react';
2 import { supabase } from '../lib/supabase';
3 import type { ContractWorker } from '../lib/supabase';
4
5 interface AuthContextType {
6   user: ContractWorker | null;
7   loading: boolean;
8   signIn: (email: string, password: string) => Promise<any>;
9   signUp: () => Promise<any>;
10 }
11
12 const AuthContext = createContext<AuthContextType> | undefined(undefined);
13
14 export const AuthProvider = ({ children }: { children: ReactNode }) => {
15   const [user, setUser] = useState<ContractWorker | null>(null);
16   const [loading, setLoading] = useState(true);
17
18   useEffect(() => {
19     const checker = () => {
20       try {
21         const storedUser = localStorage.getItem('contract_worker');
22         if (storedUser) {
23           const userData = JSON.parse(storedUser);
24           setUser(userData);
25         }
26       } catch (error) {
27         console.error('useAuth: Error checking stored user:', error);
28         localStorage.removeItem('contract_worker');
29       }
30       setLoading(false);
31     };
32     checker();
33   }, []);
34
35   const createSession = async (userId: string) => {
36     try {
37       const { data, error } = await supabase
38         .from('worker_sessions')
39         .insert([
40           {
41             contract_worker_id: userId,
42             ip_address: await getClientIP(),
43             user_agent: navigator.userAgent
44           }
45         ])
46         .select();
47     } catch (error) {
48       console.error('useAuth: Error creating session:', error);
49     }
50   };
51 }
```

Dashboard:



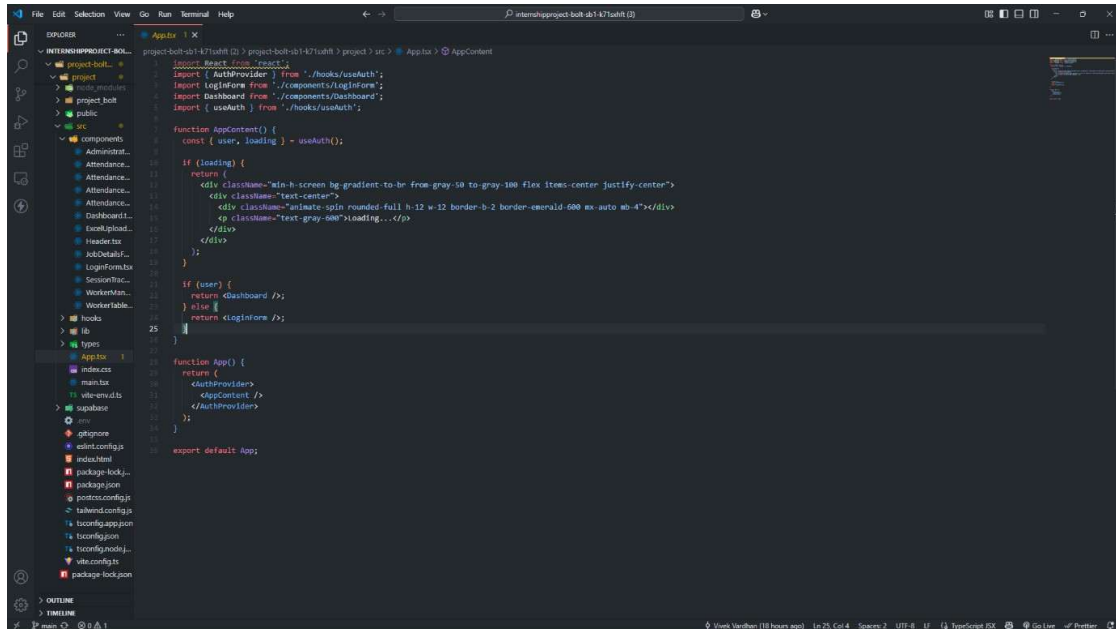
```
1 import React, { useState } from 'react';
2 import { Logout, Users, Calendar, BarChart3, Settings, Clock } from 'lucide-react';
3 import { useAuth } from '../hooks/useAuth';
4 import WorkerManagement from './WorkerManagement';
5 import AttendanceManagement from './AttendanceManagement';
6 import AttendanceReports from './AttendanceReports';
7 import SessionTracker from './SessionTracker';
8 import Administrator from './Administrator';
9
10 type TabType = 'workers' | 'attendance' | 'reports' | 'sessions' | 'admin';
11
12 const Dashboard: React.FC = () => {
13   const [activeTab, setActiveTab] = useState<TabType>('workers');
14   const { user, signOut } = useAuth();
15
16   const handleSignOut = async () => {
17     // No loading state needed since it's instant
18     await signOut();
19   };
20
21   const tabs = [
22     { id: 'workers' as TabType, label: 'Worker Management', icon: Users },
23     { id: 'attendance' as TabType, label: 'Mark Attendance', icon: Calendar },
24     { id: 'reports' as TabType, label: 'Reports', icon: BarChart3 },
25     { id: 'sessions' as TabType, label: 'Session Tracking', icon: Clock },
26     { id: 'admin' as TabType, label: 'Administrator', icon: Settings }
27   ];
28
29   return (
30     <div className="min-h-screen bg-gradient-to-br from-gray-50 to-gray-100">
31       <div className="bg-gradient-to-r from-emerald-600 to-emerald-500 shadow-lg">
32         <div className="flex items-center justify-between h-16">
33           <div className="flex shrink-0">
34             <div className="h-10 w-10 text-white">
35               <img alt="Contract Worker Portal logo" data-bbox="10 10 10 10"/>
36             <div className="text-emerald-100 text-sm">
37               Welcome, {user?.full_name} {user?.department}
38             </div>
39           </div>
40           <div>
41             <button
42               onClick={handleSignOut}
43               className="flex items-center space-x-2 px-4 py-2 bg-emerald-700 text-white rounded-lg hover:bg-emerald-800 transition-colors">
44               <Logout/>
45             </button>
46           </div>
47         </div>
48       </div>
49       <div>
50         <div>
51           {tabs.map((tab) => (
52             <div
53               key={tab.id}
54               className={activeTab === tab.id ? 'text-emerald-600' : 'text-gray-500'}>
55               <tab.icon/> {tab.label}
56             </div>
57           ))}
58         </div>
59         <div>
60           {tabs.map((tab) => (
61             <div
62               key={tab.id}
63               className={activeTab === tab.id ? 'bg-white' : 'bg-gray-50'}>
64               <tab.component/>
65             </div>
66           ))}
67         </div>
68       </div>
69     </div>
70   );
71 }
```

Admin:



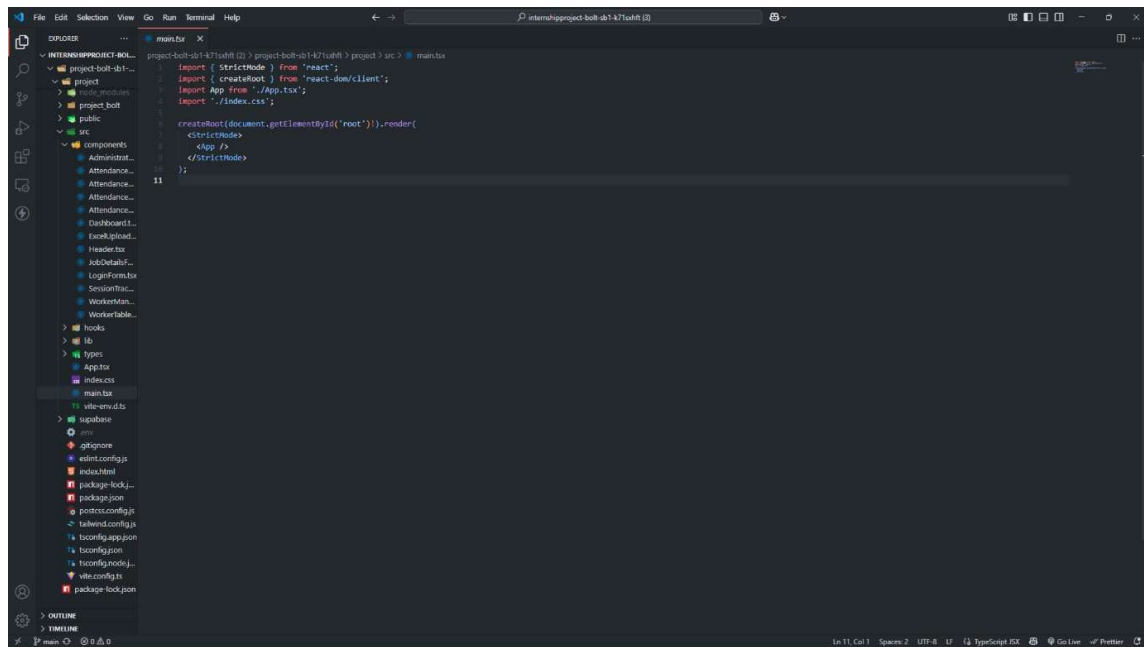
```
1 import React, { useEffect, useState } from 'react';
2 import { supabase, Contractor, Worker, AttendanceRecord } from '../lib/supabase';
3
4 const Administrator = React.FC => {
5   // const fetchContractors = () => Promisevoid
6   const fetchContractors = async () => {
7     const { data, error } = await supabase
8       .from('contract_workers')
9       .select('*')
10       .eq('is_active', true)
11       .order('full_name');
12     if (error || !data) setContractors(data);
13   };
14   fetchContractors();
15   // const fetchWorkers = () => Promisevoid
16   const fetchWorkers = async () => {
17     const { data, error } = await supabase
18       .from('workers')
19       .select('*')
20       .eq('contract_worker_id', selectedContractor)
21       .eq('is_active', true)
22       .order('worker_name');
23     setWorkers(workersData || []);
24   };
25   fetchWorkers();
26   // const fetchAttendance = () => Promisevoid
27   const fetchAttendance = async () => {
28     const { data, error } = await supabase
29       .from('attendance_records')
30       .select('*')
31       .eq('contract_worker_id', selectedContractor)
32       .order('attendance_date', { ascending: false });
33     setAttendance(attendanceData || []);
34   };
35   fetchAttendance();
36   return (
37     <div className="bg-white rounded-xl shadow-lg p-8">
38       <h2 className="text-2xl font-bold text-gray-900 mb-6">Administrator Panel</h2>
39     </div>
40   );
41 }
42 export default Administrator;
```

Application:



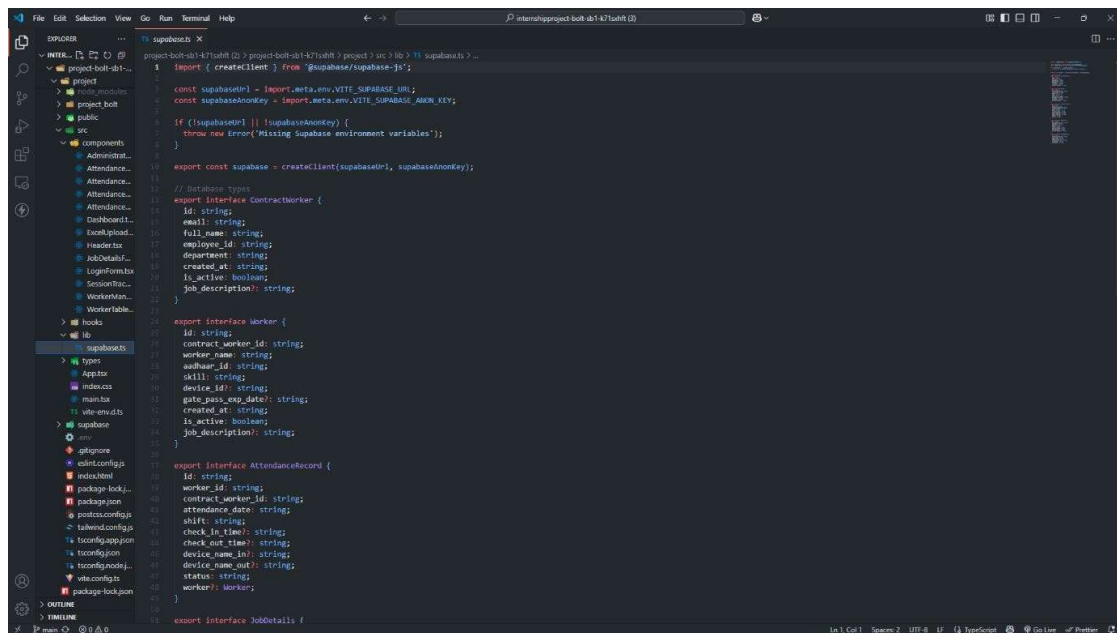
```
1 import React from 'react';
2 import { AuthProvider } from './hooks/useAuth';
3 import { LoginInfo } from './components/LoginInfo';
4 import { Dashboard } from './components/Dashboard';
5 import { useAuth } from './hooks/useAuth';
6
7 function AppContent() {
8   const { user, loading } = useAuth();
9
10   if (loading) {
11     return (
12       <div className="min-h-screen bg-gradient-to-b from-gray-10 to-gray-100 flex items-center justify-center">
13         <div className="text-center">
14           <div className="w-12 h-12 border-2 border-emerald-600 rounded-full animate-spin"></div>
15           <p className="text-gray-500">loading...</p>
16         </div>
17       </div>
18     );
19   }
20
21   if (user) {
22     return <Dashboard />;
23   } else {
24     return <LoginInfo />;
25   }
26 }
27
28 function App() {
29   return (
30     <AuthProvider>
31       <AppContent />
32     </AuthProvider>
33   );
34 }
35
36 export default App;
```

Main:



```
1  import { StrictMode } from 'react';
2  import { createRoot } from 'react-dom/client';
3  import App from './App.tsx';
4  import './index.css';
5
6  createRoot(document.getElementById('root')).render(
7    <StrictMode>
8      <App />
9    </StrictMode>
10 );
11
```

Database:



```
1  import { createClient } from '@supabase/supabase-js';
2
3  const supabaseUrl = import.meta.env.VITE_SUPABASE_URL;
4  const supabaseKey = import.meta.env.VITE_SUPABASE_ANON_KEY;
5
6  if (!supabaseUrl || !supabaseKey) {
7    throw new Error('Missing Supabase environment variables');
8  }
9
10 export const supabase = createClient(supabaseUrl, supabaseKey);
11
12 // Database types
13 export interface ContractWorker {
14   id: string;
15   email: string;
16   full_name: string;
17   employee_id: string;
18   department: string;
19   created_at: string;
20   is_active: boolean;
21   job_description: string;
22 }
23
24 export interface Worker {
25   id: string;
26   contract_worker_id: string;
27   worker_name: string;
28   aadhaar_id: string;
29   skill: string;
30   device_id: string;
31   gate_pass_exp_date: string;
32   created_at: string;
33   is_active: boolean;
34   job_description: string;
35 }
36
37 export interface AttendanceRecord {
38   id: string;
39   worker_id: string;
40   contract_worker_id: string;
41   attendance_date: string;
42   shift: string;
43   check_in_time: string;
44   check_out_time: string;
45   device_name_in: string;
46   device_name_out: string;
47   status: string;
48   worker?: Worker;
49 }
50
51 export interface JobDetails {
```

CONCLUSION

The “Gate Pass Management System” project, as evidenced by the provided screenshots, **successfully implements a digital solution for controlling and tracking personnel within a facility.** It effectively integrates a “Workers Info” module for managing detailed individual profiles (including IDs, contact, and contractor affiliations) with an “Attendance Page” that facilitates both manual and bulk (via Excel upload) recording of in/out times and device usage. This comprehensive approach enhances security through improved record-keeping and boosts operational efficiency by automating key aspects of personnel tracking, ultimately providing a transparent and accessible system for monitoring who is present and when.

REFERENCES

<https://nodejs.org/en/docs/>

<https://expressjs.com/>

<https://dev.mysql.com/doc/>

<https://www.w3schools.com/>

GLOSSARY

HTML

CSS

JAVASCRIPT

NODE JS

EXPRESS JS

MYSQL

All the above mentioned are the names of different web scripting languages used in the development.